

Out[26]: 140737266115992

```
In [27]: y=10
         y
```

Out[27]: 10

```
In [28]: id(y)
```

Out[28]: 140737266115992

```
In [29]: z=20
         z
```

Out[29]: 20

```
In [30]: id(z)
```

Out[30]: 140737266116312

Functions

```
In [31]: cars()
         german_cars('audi','bmw')
         asian_cars('Tata','Toyota','Mahindra')
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[31], line 1
----> 1 cars()

NameError: name 'cars' is not defined
```

24th

```
In [1]: i=5
         i
```

Out[1]: 5

```
In [2]: type(i)
```

Out[2]: int

```
In [3]: type()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[3], line 1
----> 1 type()

TypeError: type() takes 1 or 3 arguments
```

```
In [4]: type
```

Out[4]: type

In [5]: `type(i)`

Out[5]: int

In [6]: `f=3.4`
`f`

Out[6]: 3.4

In [7]: `type(f)`

Out[7]: float

In [8]: `petrol=110.34`
`petrol`

Out[8]: 110.34

In [9]: `gold=127000.67`
`gold`

Out[9]: 127000.67

In [10]: `type(gold)`

Out[10]: float

In [11]: `type(petrol)`

Out[11]: float

In [12]: `2e0` *#Exponential form*

Out[12]: 2.0

In [13]: `2e1` *#e0=1.0 e1=10.0 e2=100.0 e3=1000.0*

Out[13]: 20.0

In [14]: `2E1`

Out[14]: 20.0

In [15]: `3E3`

Out[15]: 3000.0

In [16]: `type(3E3)`

Out[16]: float

In [17]: `4a4` *#In float only e letter is allowed*

```
Cell In[17], line 1
  4a4
    ^
SyntaxError: invalid decimal literal
```

INT AND FLOAT IS COMPLETED

Boolean

```
In [18]: b=true      #Capital T is allowd in true
        b
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[18], line 1
----> 1 b=true
      2 b

NameError: name 'true' is not defined
```

```
In [19]: b=True
        b
```

Out[19]: True

```
In [20]: b1=False
        b1
```

Out[20]: False

```
In [21]: b2=None     #No output is displayed
        b2
```

```
In [22]: True+True
```

Out[22]: 2

```
In [23]: True-False
```

Out[23]: 1

```
In [24]: False-True
```

Out[24]: -1

```
In [25]: True*True-False
```

Out[25]: 1

```
In [26]: False/True   # Float division
```

Out[26]: 0.0

```
In [27]: False//True
```

```
Out[27]: 0
```

```
In [29]: True/False      #ZeroDivisionError
```

```
-----  
ZeroDivisionError                                Traceback (most recent call last)  
Cell In[29], line 1  
----> 1 True/False      #ZeroDivisionError  
  
ZeroDivisionError: division by zero
```

```
In [31]: True+True  
         True*False  
         True-True+True
```

```
Out[31]: 1
```

```
In [32]: print(True+True)  
         print(True*False)  
         print(True-True+True)
```

```
2  
0  
1
```

Boolean is Completed

Complex Data Types

```
In [37]: c=10+20j  
         c
```

```
Out[37]: (10+20j)
```

```
In [38]: type(c)
```

```
Out[38]: complex
```

```
In [39]: c
```

```
Out[39]: (10+20j)
```

```
In [40]: c.real
```

```
Out[40]: 10.0
```

```
In [42]: c.imag
```

```
Out[42]: 20.0
```

```
In [43]: d=20+30j
```

```
d
```

```
Out[43]: (20+30j)
```

```
In [44]: print(c)
         print(d)
```

```
(10+20j)
```

```
(20+30j)
```

```
In [45]: print(c+d)
```

```
(30+50j)
```

```
In [46]: c2=10+20z
         c2
```

```
Cell In[46], line 1
      c2=10+20z
           ^
SyntaxError: invalid decimal literal
```

```
In [47]: c3=10+20j
         c3
```

```
Out[47]: (10+20j)
```

Complex is completed

String Started !!

```
In [48]: 'Welcome to nit'
```

```
Out[48]: 'Welcome to nit'
```

```
In [49]: s='Welcome to nit'
         s
```

```
Out[49]: 'Welcome to nit'
```

```
In [50]: type(s)
```

```
Out[50]: str
```

```
In [51]: s1="Welcome to nit"
         s1
```

```
Out[51]: 'Welcome to nit'
```

```
In [52]: type(s1)
```

```
Out[52]: str
```

```
In [53]: s2='''Welcome to nit'''
```

```
s2
```

```
Out[53]: 'Welcome to nit'
```

```
In [54]: type(s2)
```

```
Out[54]: str
```

```
In [55]: s3='Welcome
         to
         nit '
```

```
Cell In[55], line 1
      s3='Welcome
      ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
In [56]: s3="Welcome
         to
         nit "
```

```
Cell In[56], line 1
      s3="Welcome
      ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
In [57]: s3='''Welcome
         to
         nit '''
s3
```

```
Out[57]: 'Welcome\n          to\n          nit '
```

```
In [58]: s4='''For Tambaram today (29 September), rain is not very likely in the next few
            \n small chance of rain around 4-6 PM,
            \ndecreasing afterward.'''
s4
```

```
Out[58]: 'For Tambaram today (29 September), rain is not very likely in the next few hou
rs. The current forecast shows partly sunny conditions, with only a \n          \n
small chance of rain around 4-6 PM, \n          \ndecreasing afterward.'
```

String indexing

```
In [59]: s
```

```
Out[59]: 'Welcome to nit'
```

```
In [60]: s[0]
```

```
Out[60]: 'W'
```

```
In [61]: s[-1]
```

```
Out[61]: 't'
```

In [62]: s[6]

Out[62]: 'e'

In []: s[]

In [63]: s[-4]

Out[63]: ' '

In []:

In []:

In []:

In []:

In []:

In []:

In []:

In []:

In []:

In []: